

Questions 1 to 4 carry 3 marks each

Questions 5 to 8 carry 4 marks each

Questions 9 to 10 carry 6 marks each

Questions 11 to 12 carry 7 marks each

Q1.

At what time will the hour hand and the minute hand be 10° apart between 4 and 5 pm?

Q2.

Write two advantages and one disadvantage of an Edge Triggered Flip-Flop over a Level Triggered Latch.

Q3.

What would be more efficient to do? Support your answer with reasons.

a) Encrypt the data first and then compress it.

b) Compress the data first and then encrypt it.

Q4.

Give the binary equivalent of: -31.275.

Q5. <<I am not sure about this but it was something of this sort>>

The design chip has an area of 81mm^2 and consumes 1mW power at a frequency of 500 kHz . What power will it consume at 1000 kHz ? Also what will be the power which the chip can sustain?

Q6.

Which gate out of an AND or a XOR will consume more power, given that they have equal transition probabilities?

Q7.

Write a function "IsLittleEndian()" which returns True if a n-bit processor is Little Endian and False if the processor is Big Endian.

Q8.

Propose a hardware scheme to square a 4-bit number.

Q9.

You have an infinite supply of Half and Full Adders. Draw a circuit which counts the number of one's present in a 7 bit number and gives the output as a 3 bit number.

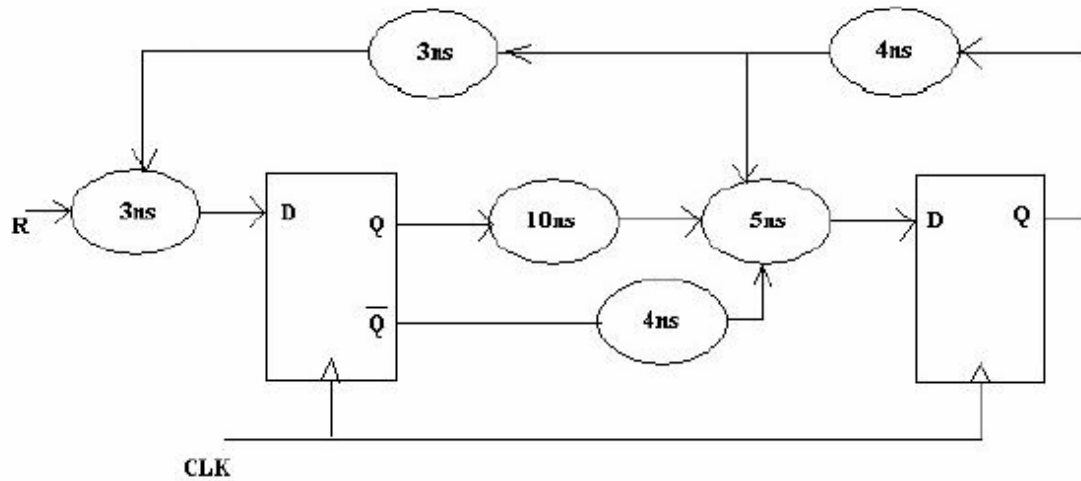
Q10.

Design a state machine which calculates the minimum number of steps a robot has to take in going from a co-ordinate (x_0, y_0) to (x_1, y_1) . The initial co-ordinates are $(0, 0)$ and the input to the machine is (x_1, y_1) . For the next input (x_2, y_2) , (x_1, y_1) is taken as the current position of the robot.

Q11. <<It was something of this sort. This circuit and data can be wrong>>
Consider the following circuit with the elements depicting their propagation delays:

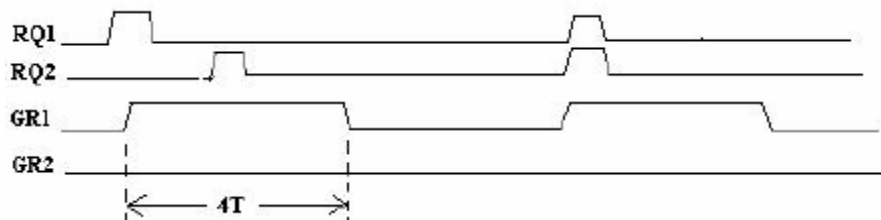
- Each Flip-Flop has a Data Setup Time of 3ns and zero t_{pd} .
- The Clock Latency (time after which it reaches the flip-flop2) = 2ns.

a) Find the maximum frequency of input R.
b) Find the delay of a delay element that can be inserted in the clock path of 2nd flip-flop if the clock rate is 125 kHz and the input is coming at a rate calculated in part a).



Q12.

Design a state machine which receives two inputs RQ1 and RQ2 and has two outputs GR1 and GR2 following the adjoining timing diagram. When a RQ pulse comes; the corresponding GR becomes high and stays there until 4T-states from it. When a GR is high, the other RQ is not served if it comes during that duration. If both RQ are asserted simultaneously, that RQ is served which has most recently been served.



Don't spend too much time in making proper designs. People here got away with making block diagrams for Q 8, 9 and 10. Rest one can easily handle.
All the Best!!!